

Neural Network Problems → Solutions

Quick Cheat Sheet

1. Vanishing Gradients

Solution
■ Use ReLU / Leaky ReLU / ELU
■ Use suitable weight initialization (e.g. He initialization with ReLU)
■ Use Batch Normalization
■ Use Residual / Skip Connections
■ Avoid very deep plain networks when unnecessary

2. Overfitting

Solution
■ Reduce model complexity
■ Increase training data / use data augmentation
■ Use Dropout
■ Use L1 / L2 regularization
■ Use Early Stopping
■ Use Cross-Validation when appropriate

3. Normalization

Solution
■ Normalize / standardize input features
■ Use Batch Normalization between layers when appropriate
■ Use Layer Normalization or Group Normalization when appropriate
■ Keep feature scales comparable

4. Exploding Gradients

Solution
■ Use Gradient Clipping
■ Use suitable weight initialization
■ Use an appropriate learning rate
■ Use Batch Normalization when appropriate

5. Optimizer Problems

Solution
■ Try Adam as a strong general-purpose starting point
■ Use SGD + Momentum when appropriate
■ Use Adagrad for sparse-feature problems
■ Use RMSprop for some recurrent/sequence models

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| ■ Tune the learning rate |
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6. Learning Rate Problems

Solution
■ Use Learning Rate Scheduling
■ Use Step Decay or Exponential Decay
■ Use ReduceLROnPlateau when validation performance plateaus
■ Use Cosine Annealing when appropriate
■ Start with a reasonable learning rate and tune it

7. Hyperparameter Tuning

Solution
■ Tune the number of hidden layers
■ Tune the number of nodes per layer
■ Tune batch size
■ Tune learning rate
■ Tune dropout rate
■ Tune activation function
■ Tune optimizer
■ Tune weight decay / L2 regularization
■ Use Grid Search, Random Search, or Optuna
■ Evaluate using validation data / cross-validation